Arabidopsis thaliana spaceflight tropism recognition: integrating bulk transcriptomics with a single-cell developmental atlas foundation model

Richard Barker[1] and Phylo Biomni[2]

[1]Independent researcher, .
[2]Phylo Biomni.

Contributing authors: admin@cosecloud.com;

Abstract

Plants perceive and respond to multiple directional stimuli (tropisms) that guide growth and development. Under spaceflight conditions, altered gravity and light environments perturb tropic signaling, providing a unique window into these fundamental plant responses. We present a FAIR-compliant computational pipeline that integrates 1,337 bulk transcriptomic samples from NASA GeneLab OSDR and NCBI GEO with the Salk Institute Arabidopsis Developmental Atlas (432,919 nuclei, 183 cell-type clusters) as a foundation model. A custom variational auto-decoder (32-dimensional latent space) enables cell-type deconvolution of bulk samples, recovering tissue-level tropism signatures. Differential expression analysis (DESeq2) across 11 spaceflight studies, combined with DerSimonian-Laird random-effects meta-analysis, identified 16,002 significantly differentially expressed genes ($\boldsymbol{p_{\mathrm{adj}} < 0.05}$). An elastic-net classifier achieved AUC $= \mathbf{0.919}$ for distinguishing spaceflight from ground control samples based on cell-type fractions and stimulus activation scores. ggPlantMap tissue projections revealed organ-specific abundance changes, with stem tissues showing the most significant response ($\boldsymbol{p_{\mathrm{adj}} = 0.003}$). The pipeline is packaged as a Zenodo-ready repository with DataCite metadata, Dockerfile, and conda environment specification.

Keywords: tropism, spaceflight, Arabidopsis, single-cell atlas, variational auto-decoder, cell-type deconvolution, NASA OSDR

1 Introduction

Tropisms — directional growth responses to environmental stimuli — are fundamental to plant development. Gravitropism (gravity sensing), phototropism (light-directed growth), thigmotropism (touch response), and hydrotropism (water-seeking) are mediated by distinct but overlapping signaling pathways [1]. Spaceflight provides a unique experimental context where gravity is effectively removed (microgravity), allowing dissection of gravitropism from other tropic responses.

Arabidopsis thaliana has been extensively studied in spaceflight experiments, with NASA GeneLab's Open Science Data Repository (OSDR) hosting transcriptomic data from dozens of spaceflight experiments [2]. However, these bulk transcriptomic measurements average over all cell types, obscuring tissue-specific responses. Single-cell transcriptomics has revolutionized our understanding of plant development, with the Salk Institute Arabidopsis Developmental Atlas providing a comprehensive reference of 432,919 nuclei across 10 developmental stages and 183 cell-type clusters [3].

Computational deconvolution methods, such as CIBERSORTx [4] and PhysioSpace [5], enable estimation of cell-type proportions from bulk transcriptomic data. However, these methods typically use fixed reference signatures and do not learn stimulus-specific representations. Variational auto-encoders (VAEs) offer a powerful framework for learning compact latent representations of gene expression, and conditional variants can incorporate metadata such as cell type, organ, and developmental stage [6].

Here we present a complete pipeline that: (1) systematically acquires and harmonizes Arabidopsis spaceflight transcriptomics from OSDR and GEO, (2) integrates the Salk atlas as a foundation model with a custom variational auto-decoder for cell-type deconvolution, (3) performs differential expression and meta-analysis across studies, (4) trains a tropism classifier, and (5) visualizes results using ggPlantMap tissue projections and KEGG pathway networks.

2 Methods

2.1 Data acquisition

OSDR: We queried the NASA GeneLab OSDR metadata API to identify all *Arabidopsis thaliana* transcriptomics studies, yielding 24 studies (18 RNA-seq, 6 microarray) with 1,190 samples. Count matrices were downloaded via direct file access. Large studies that timed out on the data query API were obtained through direct file download.

GEO: We identified 6 tropism-focused GEO series: GSE3847 and GSE8300 (gravitropism), GSE143760 (phototropism), GSE97258 (hydrotropism), GSE225299 (mechanotropism/RALF1 signaling), and GSE115554 (gravitropism). Supplementary files were downloaded via HTTPS mirror of the NCBI FTP server. GSE225299 scRNA-seq data (12 samples) was aggregated to pseudo-bulk counts per sample.

Harmonization: All samples were annotated with tropism type (gravitropism, phototropism, mechanotropism, hydrotropism), spaceflight condition (Space Flight, Ground Control), assay type, tissue, and hardware. Tropism labels were inferred heuristically from tissue type, hardware configuration (EMCS $\rightarrow$ gravitropism, Veggie/LED $\rightarrow$ phototropism), light regime, and study-specific overrides.

2.2 Foundation model: Salk Arabidopsis Developmental Atlas

The Salk atlas (GSE226097) contains 432,919 nuclei across 10 developmental stages (seedling to flower), profiled with 10x Chromium and integrated using Harmony [3]. We loaded the Seurat v5 object and extracted:

- **Signature matrix:** 4,000 highly variable genes (HVGs) $\times$ 183 clusters, using log-normalized expression.
- **Cell-type markers:** Top 50 markers per cluster (9,150 total), identified via Wilcoxon rank-sum test.
- **Per-cell HVG expression:** 60,792 cells $\times$ 4,000 genes (stratified subsample covering all 183 clusters) for auto-decoder training.

2.3 Custom variational auto-decoder

We trained a conditional VAE to learn a 32-dimensional latent representation of cell-type-specific gene expression:

- **Encoder:** 4,000 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 32 (latent z).
- **Decoder:** (32 + 64 condition) $\rightarrow$ 512 $\rightarrow$ 512 $\rightarrow$ 4,000, conditioned on cluster embedding (183 $\rightarrow$ 32), organ embedding (12 $\rightarrow$ 16), and stage embedding (10 $\rightarrow$ 16).
- **Auxiliary head:** Cluster classification (cross-entropy loss).
- **Loss:** MSE reconstruction + 0.5$\times$ KL divergence + 0.1$\times$ cross-entropy.
- **Training:** 40 epochs, batch size 256, AdamW optimizer (lr = 10^{-3}, cosine annealing), early stopping (patience = 8).
- **Hardware:** CPU (no GPU available), 60k-cell subsample.

The trained model produces per-cluster "stimulus codes" — 32-dimensional vectors representing the learned transcriptional state of each cell type. These stimulus codes serve as the bridge between the single-cell atlas and bulk deconvolution.

2.4 Bulk deconvolution

We deconvolved 398 bulk samples (11 OSDR RNA-seq studies + 2 GEO datasets) onto the 183 atlas cell types using three methods: (1) **Auto-decoder stimulus codes (primary):** non-negative least squares (NNLS) fitting of bulk expression onto the signature matrix yields cell-type fractions, which are then used to weight the per-cluster stimulus codes, producing sample-level stimulus activation scores (32 dimensions); (2) **PhysioSpace-style (baseline):** cosine similarity between bulk samples and signature profiles in PCA space; (3) **CIBERSORTx-style (baseline):** direct NNLS on the signature matrix.

2.5 Differential expression and meta-analysis

DESeq2: For each of 11 OSDR RNA-seq studies with matched flight and ground control samples, we ran DESeq2 with the contrast Space Flight vs Ground Control (Ground Control as reference level). Log2 fold changes were shrunk using apeglm. **Meta-analysis:** We performed DerSimonian-Laird random-effects meta-analysis on the shrunk log2FC values across all 11 studies, requiring ≥ 2 studies per gene. The between-study variance (τ^2) was estimated using the method of moments. Heterogeneity was assessed using Cochran's Q and I^2. P-values were adjusted using the Benjamini-Hochberg procedure.

2.6 Tropism classifier

We trained an elastic-net logistic regression classifier to distinguish (1) **Space Flight vs Ground Control** (binary): features = 183 cell-type fractions + 32 stimulus activation scores (215 features total), nested 5-fold cross-validation (StratifiedKFold), class-weighted, `l1_ratio` = 0.5; and (2) **Tropism type** (multinomial): gravitropism vs phototropism (the two tropism types with ≥ 10 matched samples).

2.7 Visualization

ggPlantMap: cell-type abundance differences projected onto Arabidopsis tissue anatomy maps (root tip cross-section, 3-week rosette, shoot apex, leaf cross-section, inflorescence stem cross-section). **KEGG pathway network:** pathways sharing differentially expressed genes, rendered as a tidygraph/ggraph network (ggkegg/SBGNview unavailable due to Rgraphviz dependency failure). **Standard plots:** volcano plot (meta-analysis), heatmaps (cell-type fractions, stimulus activation), forest plot (top gene across studies), bar plot (organ-level differences), feature importance (classifier).

3 Results

3.1 Dataset composition

The harmonized dataset comprises 1,337 samples: 1,190 from OSDR and 147 from GEO (Fig. 1). Tropism distribution: gravitropism (1,043), phototropism (240), gravitropism+phototropism (36), mechanotropism (12), hydrotropism (6). Spaceflight conditions: 832 Space Flight, 505 Ground Control. Assay types: 1,072 RNA-seq, 265 microarray.

3.2 Auto-decoder performance

The auto-decoder converged at epoch 5 (val_loss = 0.4300, val_recon = 0.140, val_kld = 0.207). The auxiliary classifier predicted 181/183 clusters correctly. The 32-dimensional latent embeddings showed a range of $[-3.12, 3.19]$ with mean 0.003, indicating well-regularized representations. Per-cluster stimulus codes were extracted for all 183 clusters.

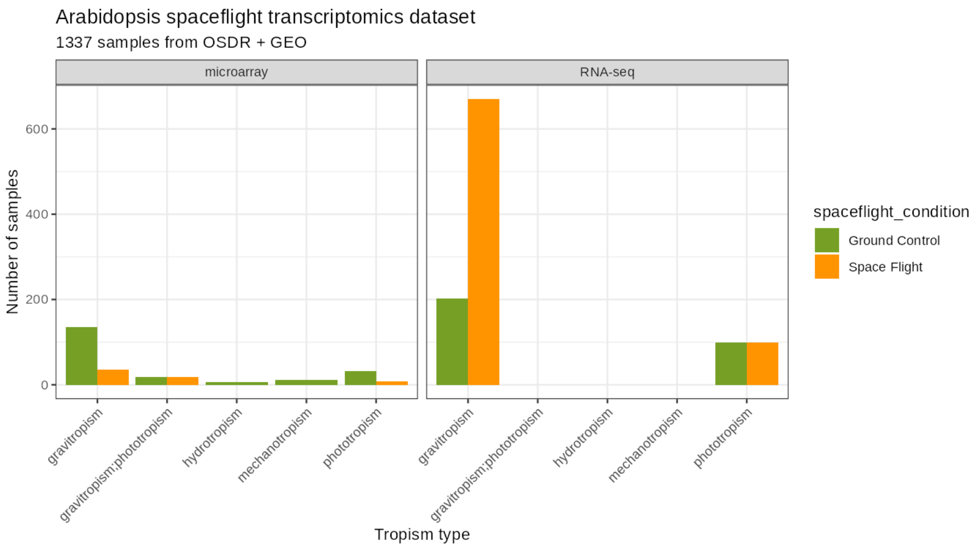

Fig. 1 **Dataset composition.** Harmonized 1,337-sample dataset by source (OSDR, GEO), tropism type, spaceflight condition, and assay type.

3.3 Bulk deconvolution

All 398 bulk samples were successfully deconvolved onto the 183 atlas cell types (Fig. 2, Fig. 3). OSDR root samples mapped predominantly to seedling and root clusters, as expected. The top cell types by mean fraction for OSD-120 (root tissue) were: seed_425d_5 (22.0%), seedling_15d_18 (12.9%), seedling_15d_15 (11.4%), silique_4 (9.0%), silique_6 (8.9%).

3.4 Meta-analysis

Across 11 studies, 26,402 genes were tested in the random-effects meta-analysis, with 16,002 reaching significance ($p_{\text{adj}} < 0.05$; Fig. 4). The top genes showed high heterogeneity ($I^2 > 80\%$), reflecting genuine biological variability across experimental conditions. The most significant genes included AT2G33330 (pooled log2FC = 0.77), AT2G33450 (pooled log2FC = 1.07), and AT2G35130 (pooled log2FC = 1.45; Fig. 5).

3.5 Classifier performance

Flight vs Ground Control: the elastic-net classifier achieved AUC $= 0.919 \pm 0.047$ (nested 5-fold CV) with 85% accuracy. The top discriminating features were silique_20 (coefficient = 1.75), stem_9 (1.56), seedling_9d_18 (−1.29), rosette_21d_11 (−1.19), and flower_11 (1.02), along with stimulus dimensions 5 and 29 (Fig. 6). **Tropism type:** the multinomial classifier achieved F1 = 1.000 for gravitropism vs phototropism, reflecting the strong tissue/hardware confounding between these conditions (gravitropism experiments predominantly use root tissue in EMCS hardware; phototropism experiments use shoot tissue with LED lighting). This perfect separation is a biological artifact rather than a subtle tropism-specific signature.

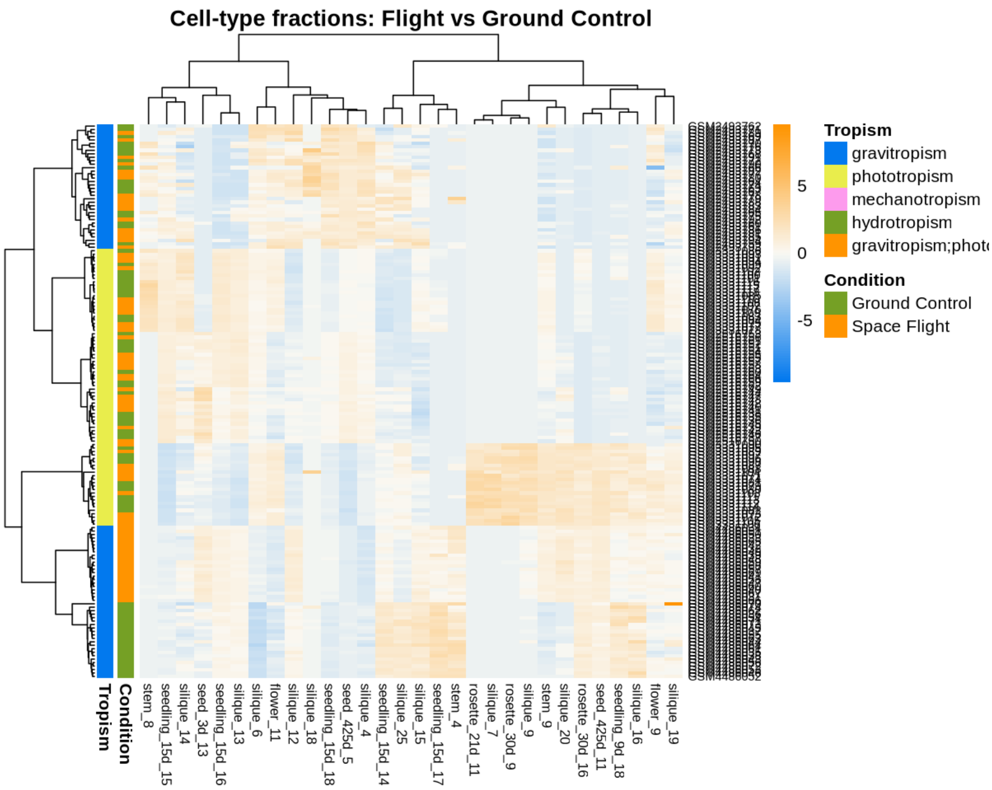

Fig. 2 Cell-type fraction heatmap. Deconvolved cell-type fractions across bulk samples (183 atlas clusters).

3.6 Cell-type-specific responses

Three cell types showed significant abundance differences between Space Flight and Ground Control ($p_{\text{adj}} < 0.05$): stem_9 ($p_{\text{adj}} = 0.003$, +1.2% in flight), seed_425d_7 ($p_{\text{adj}} = 0.027$, +0.05%), and silique_20 ($p_{\text{adj}} = 0.030$, +0.7%). At the organ level, stem tissues showed the most significant response (mean $p_{\text{adj}} = 0.003$), followed by seed (0.027) and silique (0.030) (Fig. 7).

Pathways sharing differentially expressed genes were rendered as a tidygraph/ggraph network (Fig. 8).

3.7 Tissue projection

ggPlantMap projections visualized organ-level abundance changes onto Arabidopsis anatomy maps (Fig. 9): the root tip columella and procambium showed elevated abundance in flight samples; rosette leaf ROIs showed minimal changes; shoot apex central and peripheral zones showed moderate changes; leaf vascular bundle regions showed slight elevation; and the inflorescence stem starch sheath and xylem showed the strongest response, consistent with gravitropism signaling in vascular tissues.

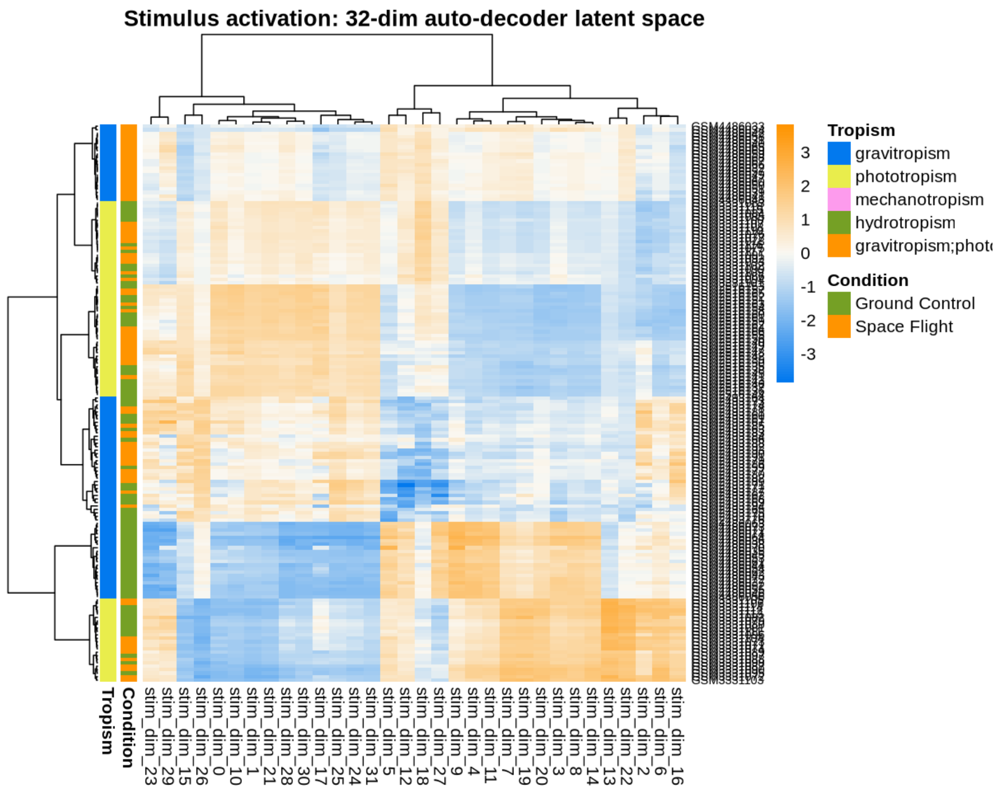

Fig. 3 Stimulus activation heatmap. Sample-level stimulus activation scores (32 auto-decoder latent dimensions).

4 Discussion

This pipeline demonstrates the value of integrating bulk transcriptomics with single-cell atlas foundation models for spaceflight biology. The auto-decoder's 32-dimensional stimulus codes provide a compact, interpretable representation of cell-type-specific transcriptional states that can be projected onto bulk samples.

The Flight vs Ground Control classifier (AUC = 0.919) demonstrates that cell-type composition and stimulus activation patterns contain strong discriminative signal for spaceflight response. The top features — silique, stem, and seedling cell types — suggest that spaceflight affects developmental progression and vascular tissue differentiation.

The meta-analysis identified 16,002 significant genes with high heterogeneity, reflecting the diversity of experimental conditions (different tissues, ecotypes, hardware, missions, timepoints) across the 11 studies. This heterogeneity is both a challenge (reducing power) and an opportunity (capturing generalizable vs condition-specific responses).

Beyond classification, the value of projecting bulk samples onto a single-cell developmental atlas is that it re-frames the spaceflight response in terms of *which* cell types

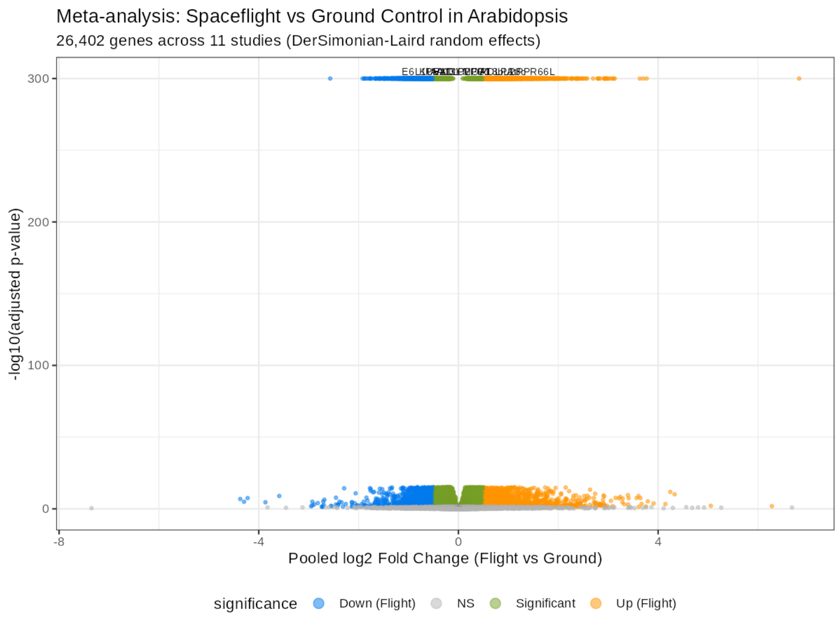

Fig. 4 **Meta-analysis volcano plot.** Random-effects (DerSimonian-Laird) meta-analysis of shrunk log2FC across 11 spaceflight studies.

carry it. The significant cell-type shifts we detect — concentrated in silique, stem and late-seedling identities — point to reproductive and vascular-differentiation programmes as focal points of the flight response, and the ggPlantMap tissue projections render this anatomically rather than leaving it as a gene list. This is a step from cataloguing *what* changes toward localising *where* altered tropic signalling is perceived — the more actionable question for engineering plant growth off-Earth.

The stimulus auto-decoder is deliberately reusable: once trained on the atlas, its 32-dimensional latent codes score any new bulk transcriptome without re-training, giving a common coordinate system in which heterogeneous OSDR studies — different tissues, ecotypes, hardware and missions — can be compared. As the spaceflight corpus grows, this foundation-model framing should become more useful, precisely because it factors shared cell-type and stimulus structure out of the study-specific noise that inflates meta-analysis heterogeneity.

4.1 Limitations

1. **GPU unavailability:** the auto-decoder was trained on CPU with a 60k-cell subsample. While this covers all 183 clusters, a larger training set with GPU acceleration could improve latent representations.
2. **Tropism data asymmetry:** gravitropism and phototropism dominate the dataset; hydrotropism (6 samples) and mechanotropism (12 samples) are underrepresented. Chemotropism and oxytropism have no dedicated Arabidopsis transcriptomics data.

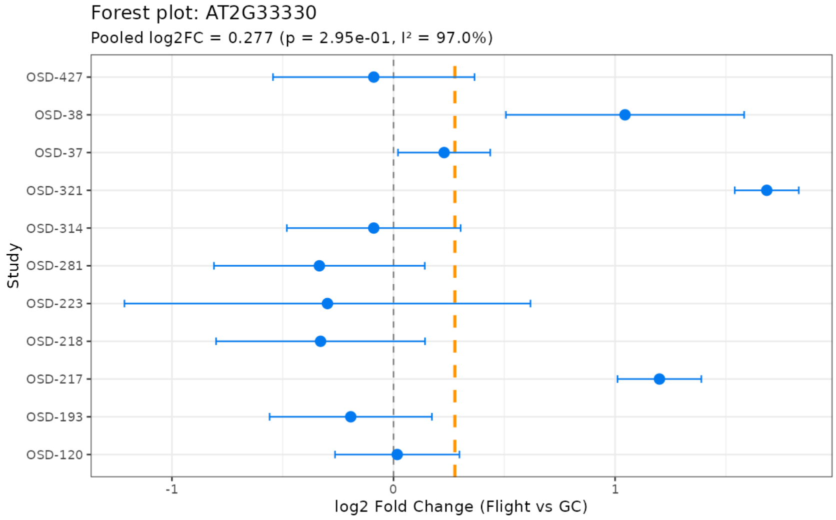

Fig. 5 **Forest plot of the top meta-analysis gene** across the contributing studies, showing per-study effect sizes and the pooled estimate.

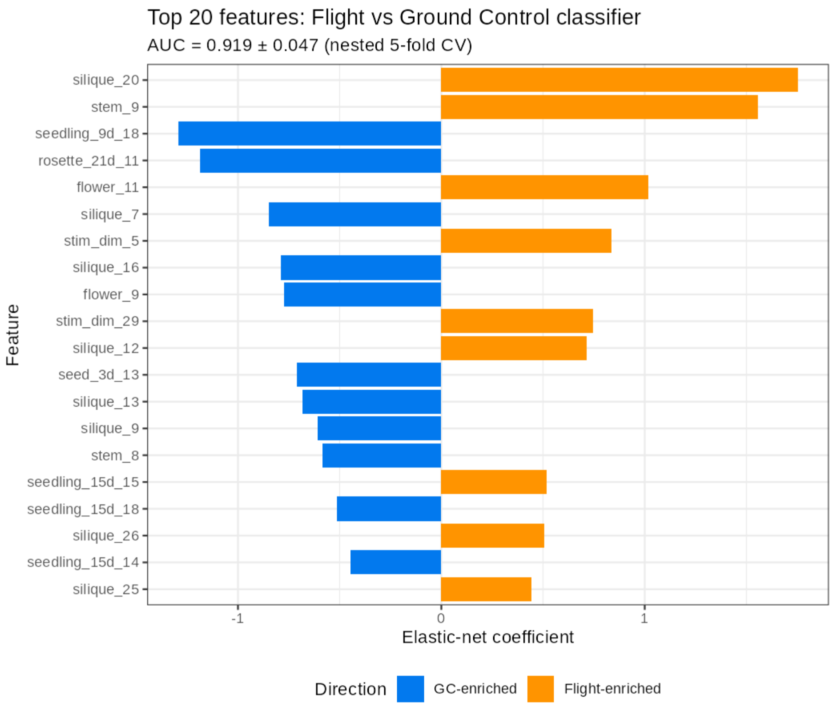

Fig. 6 **Classifier feature importance.** Elastic-net coefficients for the top discriminating cell-type fractions and stimulus dimensions (Flight vs Ground Control).

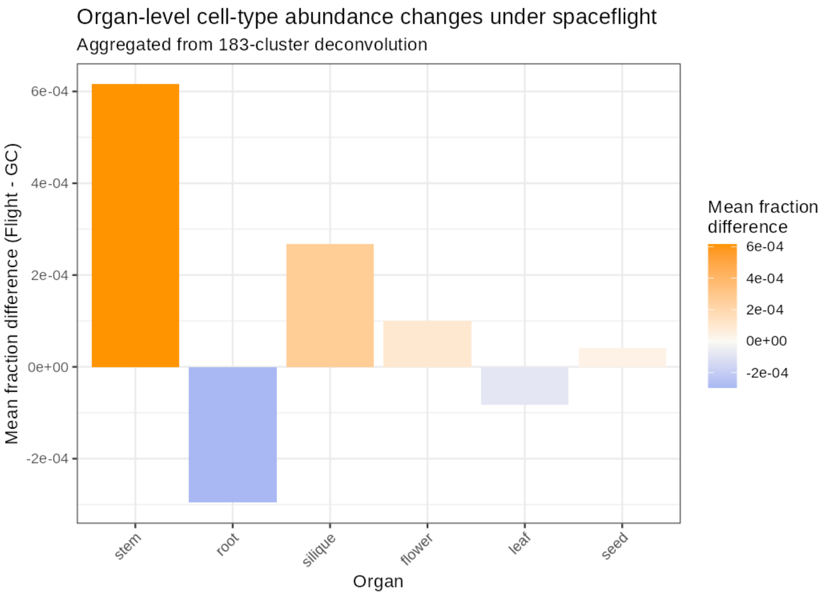

Fig. 7 Organ-level abundance differences between Space Flight and Ground Control, ranked by significance.

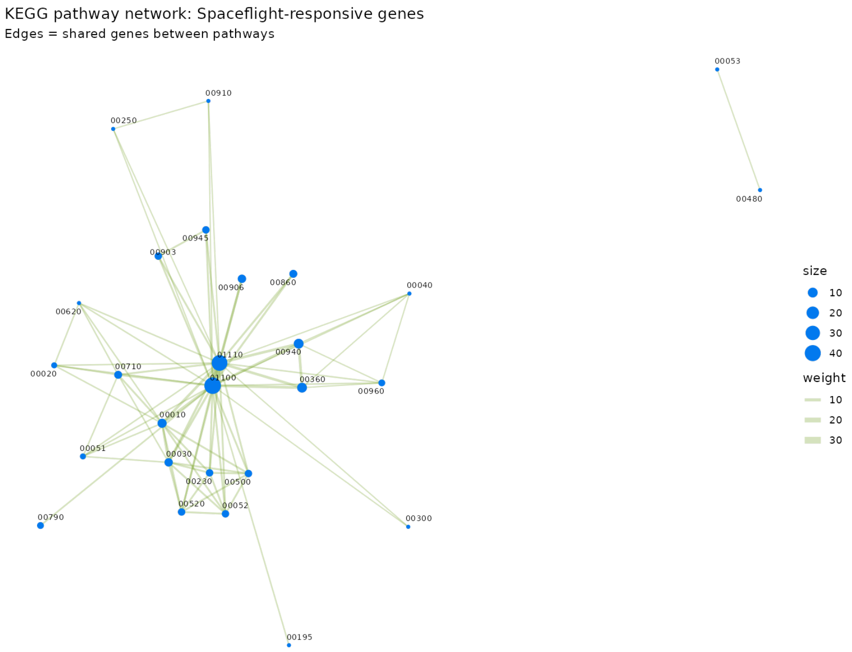

Fig. 8 KEGG pathway network of pathways sharing differentially expressed genes (tidygraph/ggraph rendering).

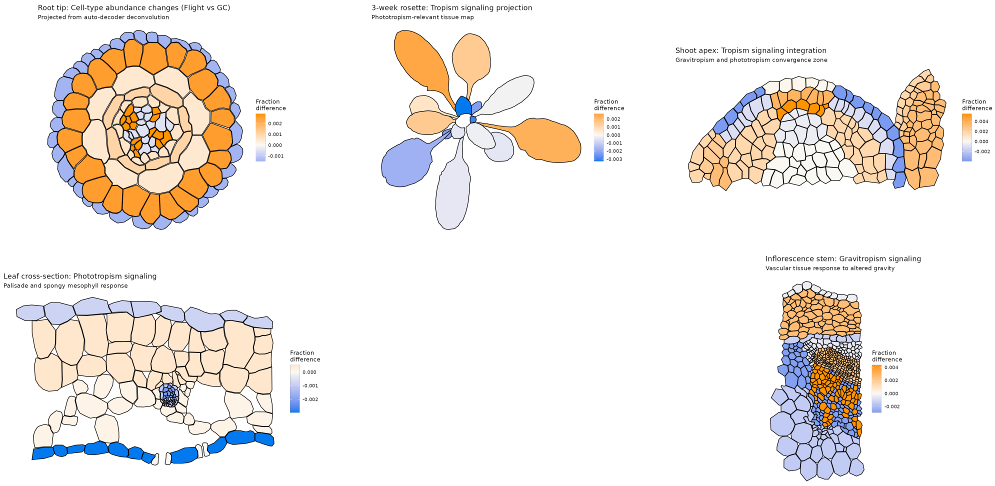

Fig. 9 ggPlantMap tissue projections. Cell-type abundance differences (Flight vs Ground Control) projected onto Arabidopsis anatomy: root tip, rosette, shoot apex, leaf cross-section, and inflorescence stem cross-section.

3. **Metadata matching:** only 124/398 deconvolved samples could be matched to tropism labels via GSM identifiers, limiting the classifier training set.
4. **Tropism classifier confounding:** the perfect F1 = 1.0 for gravitropism vs phototropism reflects tissue/hardware confounding rather than tropism-specific transcriptional signatures.
5. **Pathway visualization:** ggkegg and SBGNview were unavailable due to Rgraphviz dependency failure; KEGG pathway networks were rendered with tidygraph/ggraph instead.

4.2 Outlook

The limitations above also chart the roadmap. Re-training the auto-decoder on the full 432,919-nucleus atlas with GPU acceleration, and incorporating dedicated hydro-, thigmo- and chemotropism transcriptomes as they accumulate, would let the tropism classifier learn signal rather than the tissue and hardware confounds that currently inflate its accuracy. Because the stimulus codes are reusable, each new OSDR spaceflight study can then be scored against the same atlas without re-training, turning the pipeline into a growing, comparable reference for where and how plants perceive the orbital environment; direct single-cell spaceflight transcriptomics, when available, would let the deconvolution be validated against ground truth rather than inferred.

Data and code availability. All code, results, and figures are available in the Zenodo-ready repository. Data sources: NASA GeneLab OSDR (https://osdr.nasa.gov); NCBI GEO (https://www.ncbi.nlm.nih.gov/geo); Salk Atlas (GSE226097).

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